Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B.Tech. CE and 3rd sem
Name of the Paper: Geomatic Engineering
Paper Code: TCE-302

Time: 3 hours

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) CO1

a. Explain the principle of surveying and how it ensures accuracy in measurements.

b. A chain of 20 m is found to be 10 mm longer than its standard length. It is used to measure a plot of land where the recorded length is 240 m and breadth is 160 m. Calculate:
i. The corrected length and breadth of the plot.
ii. The corrected area of the plot in square meters.

c. Design a step-by-step procedure for conducting a theodolite survey, ensuring both temporary and permanent adjustments are considered

Q2. (2X10=20 Marks) CO2

a. Define traverse in surveying and explain two types of traversing methods.

b. In a compass survey, the bearings of four lines are recorded as follows:
i. Line AB: N30°E
ii. Line BC: S60°E
iii. Line CD: S45°W
iv. Line DA: N50°W
compute the included angles for each corner of the traverse. Verify if the traverse closes.

c. Compare triangulation and trilateration based on their principles and field applications.

Q3. (2X10=20 Marks) CO3

a. Define "contour". Explain the significance of contours in topographic mapping.

b. During reciprocal levelling between two points, the following readings were recorded:
- Instrument at A: Back sight on B = 2.375 m; Fore sight on B = 1.245 m
- Instrument at B: Back sight on A = 2.410 m; Fore sight on A = 1.210 m

Calculate the corrected difference in elevation between A and B.

c. During a fly levelling operation, the following observations were made:
BS: 0.650, 2.155, 1.405, 2.655, 2.435 m
FS: 2.455, 1.305, 0.555, 2.405 m
The first backsight was taken on a BM of RL 90.500 m. From the last back sight, it is required to set four pegs each at 30m on a falling gradient of 1 in 100, Calculate the RLs of these pegs.

Q4.

(2X10=20 Marks) CO4

a. Define Plane Table Surveying and state its principle. List four essential pieces of equipment used in plane table surveying.

b. A plane table is used for a survey, and a three-point problem arises due to inaccessible points. Illustrate the practical steps to resolve this problem in the field.

c. Create a schematic layout plan for a canal survey, considering engineering specifications.

Q5.

(2X10=20 Marks) CO5

a. Define random errors and explain their significance in surveying measurements.

b. Discuss the principle of least squares and its importance in adjusting observations.

c. The following measurements of a length are taken with their respective weights:

Observation (m)	Weight
100.5	4
100.3	2
100.6	3
100.4	1

Calculate the Most Probable Value (MPV) of the length using the weighted mean formula. Also Compute the standard error for the MPV.